

This appendix gives a minimum-fuss summary of the set-theoretic notions and facts, such as Zorn's lemma and transfinite induction, that are used in Chapter 8.

Let A, B be sets. If there exists a bijective map between A and B , we write $|A| = |B|$ and say that A and B have *the same cardinality*. This is clearly an equivalence relation between sets, and we may think of the *cardinality* $|A|$ of A as the equivalence class containing A . We write $|A| \leq |B|$ if there exists an injective map $A \rightarrow B$. This is clearly well defined, and it is a partial ordering: if there are injective maps $A \rightarrow B$ and $B \rightarrow A$, there is also a bijection $A \rightarrow B$.¹ For every set there exists another that is bigger; for example, $|A| < |B|$ when B is the power set of A , the set of all its subsets.

cardinality

The natural numbers are defined inductively as $n := \{0, \dots, n-1\}$, starting with $0 := \emptyset$. The usual expression of $|A| = n$ can then be read more formally as an abbreviation for $|A| = |n|$.

 $\mathbb{N}$

A set A is *finite* if there is a natural number n such that $|A| = n$; otherwise it is *infinite*. A is *countable* if $|A| \leq |\mathbb{N}|$, and *countably infinite* if $|A| = |\mathbb{N}|$. A bijection $\mathbb{N} \rightarrow A$ is an *enumeration* of A . If A is infinite then $|\mathbb{N}| \leq |A|$. Thus, $|\mathbb{N}|$ is the smallest infinite cardinality; it is denoted by $\aleph_0$. There is also a smallest uncountable cardinality, denoted by $\aleph_1$. If $|A| = |\mathbb{R}|$ then A is uncountable, and we say that A has *continuum many* elements. For example, there are continuum many infinite 0–1 sequences. (Whether $|\mathbb{R}|$ is equal to $\aleph_1$ or greater depends on the axioms of set theory assumed; in our context, this question does not arise.) We remark that if A is infinite and its elements are countable sets, then the union of all these sets is no bigger than A itself: $|\bigcup A| \leq |A|$.

¹ This is the Cantor-Bernstein theorem; a simple graph-theoretic proof is given in Proposition 8.4.8.

An element x of a partially ordered set X is *minimal* in X if there is no $y \in X$ with $y < x$, and *maximal* if there is no $z \in X$ with $x < z$. A partially ordered set may have one or many elements that are maximal or minimal, or none at all. An *upper bound* in X of a subset $Y \subseteq X$ is any $x \in X$ such that $y \leq x$ for all $y \in Y$.

successor A *chain* is a partially ordered set in which every two elements are comparable. If $(C, \leq)$ is a chain, and if $x, y \in C$ satisfy $x < y$ but no element z of C is such that $x < z < y$, then x is called the *predecessor* of y in C , and y the *successor* of x . A set of the form $\{x \in C \mid x < z\}$, for a given $z \in C$, is a proper *initial segment* of C .

well-ordering A partially ordered set $(X, \leq)$ is *well-founded* if every non-empty subset of X has a minimal element, and a well-founded chain is said to be *well-ordered*. For example, $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{R}$ are all chains (with their usual orderings), but only $\mathbb{N}$ is well-ordered. Note that every element x of a well-ordered set X has a successor (unless x is maximal in X): the unique minimal element of $\{y \in X \mid x < y\} \subset X$. However, an element of a well-ordered set need not have a predecessor, even if it is not minimal.

limit An element that has no predecessor is called a *limit*; for example, the number 1 is a limit in the well-ordered set

$$A = \{1 - \frac{1}{n+1} \mid n \in \mathbb{N}\} \cup \{2 - \frac{1}{n+1} \mid n \in \mathbb{N}\}$$

of rationals.

One of the many statements equivalent to the axiom of choice (which we assume throughout) is that for every set X there exists a relation by which X is well-ordered:

Well-ordering theorem. *Every set can be well-ordered.*

Two well-ordered sets are said to have *the same order type* if there is a bijection between them which preserves their orders. Thus $\mathbb{N}$ and the set of even natural numbers have the same order type, but this differs from the order type of the set A defined above. Having the same order type is clearly an equivalence relation, which justifies the term if we think of those order types themselves as equivalence classes.

ordinals When one considers properties shared by all well-ordered sets of the same order type, it is convenient to represent each order type by a specially chosen set of that type, its *ordinal*. The ordinal representing the order type of $\mathbb{N}$, for instance, is by custom denoted as ω ; our example above thus says that the set of even natural numbers has (the) order type (of) ω . Finite chains of the same cardinality always have the same order type; we choose n as the ordinal representing the chains of order n .

ω There is an *addition* of ordinals, defined by taking as the sum $\alpha + \beta$ the ordinal representing the order type of the concatenation of α with β (in this order); note that this is again a well-ordered set. For example, $\alpha + 1$ is the successor of α . Note that no inverse operation ‘ $-$ ’ is defined.

If an ordinal β has the same order type as a proper initial segment of another ordinal α , we write $\beta < \alpha$. For example, we have $0 \leq n < \omega$ for every natural number n . It can be shown that $<$ defines an ordering, even a well-ordering, on every set of ordinals. On $\mathbb{N}$, this ordering coincides with the usual one, so our notation is unambiguous.

Since a set S of ordinals is itself well-ordered, it has an order type – just like any other well-ordered set. If the ordinal α is a strict upper bound for S , then the order type of S is at most α ; it is equal to α if S consists of all the ordinals up to (but excluding) α . In fact, just like the natural numbers, infinite ordinals are usually defined in such a way that α and $\{\beta \mid \beta < \alpha\}$ are actually identical; then our ordering $<$ for ordinals coincides with the relation $\in$.

This makes it natural to write a well-ordered set S , of order type α say, as a family $S = \{s_\beta \mid \beta < \alpha\}$ with $s_\gamma < s_\beta$ for all $\gamma < \beta < \alpha$. This is common practice when one proves statements about the elements of S by *transfinite induction*, which works as follows.

*transfinite
induction*

Suppose we want to show that every $s \in S$ satisfies some proposition P ; let us write $P(s)$ to express that it does. Just as in ordinary induction we prove, for every $\beta < \alpha$, that *if* P holds for every s_γ with $\gamma < \beta$ *then* P also holds for s_β . In practice, we usually have to distinguish the two cases of β being a limit ordinal or a successor. Checking $P(s_0)$ from first principles, as in ordinary induction, is part of the first case, because 0 counts as a limit and the premise of P_γ for all $\gamma < 0$ is void. The conclusion then is that $P(s_\beta)$ for every $\beta < \alpha$, that is, every $s \in S$ satisfies P .

This is certainly simple – but is it correct? Well, any proper justification of transfinite induction requires a formal treatment of set theory, but so does ordinary induction. Informally, what we have shown is that the set

$$\{\beta < \alpha \mid P(s_\beta) \text{ fails}\}$$

has no least element. Since it is well-ordered, it must therefore be empty, so $P(s_\beta)$ holds for all $\beta < \alpha$.

Similarly, we may define things inductively. Such a *recursive definition* specifies for each ordinal α some object x_α , in a way that may refer to the objects x_β with $\beta < \alpha$ (which we think of as ‘having been defined earlier’). Our definition of the natural numbers at the start of this appendix is a simple example.

*recursive
definition*

In practice, the definition of x_α often makes sense only for ordinals α less than some fixed ordinal α^* , although the smallest such α^* may not be known in advance. For example, if the x_α are to be distinct vertices picked recursively from a graph G according to some given rules, it is clear that we shall not be able to find such x_α for all $\alpha < \alpha^*$ when $|\alpha^*| > |G|$, because $\alpha \mapsto x_\alpha$ would be an injective map from α^* to $V(G)$ showing that $|\alpha^*| \leq |G|$. Since there exist ordinals larger than $|G|$, such as any ordinal equivalent to a well-ordering of the power set of $V(G)$,

this means that our recursion cannot go on indefinitely, i.e. we shall not be able to define x_α for all ordinals α . We may not know in advance when our recursion will get stuck, i.e., which is the smallest ordinal α for which x_α cannot be found in compliance with our rules. But this does not matter: we simply *define* α^* as the first ordinal α for which x_α cannot be found, content ourselves with having defined x_α for all $\alpha < \alpha^*$, and say that our recursion *terminates* at step α^* . (In fact, we usually *want* a recursive definition to terminate. In our example, we might wish to consider the set of all vertices $x \in G$ that got picked by our definition, and this will be the set $\{x_\alpha \mid \alpha < \alpha^*\}$.)

Note that our recursive definition for x_α may involve choices. In our example, x_α might be required to be a neighbour of some x_β with $\beta < \alpha$, but there may be several such x_β , each with several neighbours that have not yet been picked. This does not cause our recursion to get stuck at step α : we just pick one eligible vertex as x_α , and proceed. In other words, we accept $\{x_\alpha \mid \alpha < \alpha^*\}$ as a properly defined set even though we may not ‘know’ its elements x_α constructively.

Back to proving things, here is a formal statement of Zorn’s lemma:

Zorn’s Lemma. *Let $(X, \leq)$ be a partially ordered set such that every chain in X has an upper bound in X . Then X contains at least one maximal element.*

Note that, in applications of Zorn’s lemma, the relation $\leq$ need not correspond to an intuitive notion of ‘smaller than’. Applied to sets or to graphs, for example, it can stand for ‘ $\supseteq$ ’ just as much as for ‘ $\subseteq$ ’. Then the ‘upper bound’ of a chain $\mathcal{C}$ is typically its overall intersection $\bigcap \mathcal{C}$.

Finally, compactness. The infinity lemma discussed in Chapter 8 generalizes as follows.² As before, we consider a collection $\{X_p \mid p \in P\}$ of finite sets, but rather than indexing these by natural numbers we have one such set X_p for every element p of some partially ordered set $(P, \leq)$. All we assume about P is that every two elements have a common upper bound: for all p, q there exists an r such that $p \leq r$ and $q \leq r$. Furthermore, we have maps $f_{qp}: X_q \rightarrow X_p$ for all $q > p$, which are compatible in that $f_{qp} \circ f_{rq} = f_{rp}$ whenever $r > q > p$.

Generalized Infinity Lemma. *For every such family $\{X_p \mid p \in P\}$ of finite sets there exists a family $\{x_p \mid p \in P\}$ of representatives $x_p \in X_p$ such that $f_{qp}(x_q) = x_p$ whenever $q > p$.*

The infinity lemma is clearly a special case of this, with $P = \mathbb{N}$ and the f_{qp} defined by iterating the lemma’s predecessor function f .

² In category theory, our ‘generalized infinity lemma’ is known as the fact that the inverse limit of any directed inverse system of finite sets is non-empty. Instead of finite sets one can take any other compact spaces; see Chapter 8.8 for an application.

The following more combinatorial encoding for compactness arguments brings out particularly well the choices involved, and their interdependence. Let X be any set, S a finite set, and $\mathcal{F}$ a set of finite subsets of X . Assume that every $Y \in \mathcal{F}$ comes with a fixed set $\mathcal{A}(Y)$ of $Y \rightarrow S$ functions, its *admissible functions*. Call $\mathcal{Y} \subseteq \mathcal{F}$ *compatible* if there exists a function $f: X \rightarrow S$ all whose restrictions to the sets in $\mathcal{Y}$ are admissible, i.e. which satisfies $f|_Y \in \mathcal{A}(Y)$ for all $Y \in \mathcal{Y}$.

Compactness Principle. $\mathcal{F}$ is compatible if every finite $\mathcal{Y} \subseteq \mathcal{F}$ is compatible.

The proofs of the generalized infinity lemma and of the compactness principle are each just a few lines based on Tychonoff's theorem that every product of compact spaces is compact. We illustrate this for the compactness principle.

Think of the set of all $X \rightarrow S$ functions as a product of $|X|$ copies of S , so they form a compact space. For every finite $Y \subseteq X$, the set of all functions $f: X \rightarrow S$ with $f|_Y \in \mathcal{A}(Y)$ is closed (as well as open) in this product space. The result now follows from the 'finite intersection property' of compact spaces, that a family of closed sets has a non-empty intersection as soon as all its finite subfamilies do.

This appendix offers a summary of background information about surfaces, as needed for an understanding of their role in the proof of the graph minor theorem or the proof of the ‘general Kuratowski theorem’ for arbitrary surfaces given in Chapter 12.7. In order to be read at a rigorous level it requires familiarity with some basic definitions of general topology (such as of the product and the identification topology), but no more.

A *surface*, for the purpose of this book, is a compact connected¹ Hausdorff topological space S in which every point has a neighbourhood homeomorphic to the Euclidean plane $\mathbb{R}^2$. An *arc*, a *circle*, and a *disc* in S are subsets that are homeomorphic in the subspace topology to the real interval $[0, 1]$, to the unit circle $S^1 = \{x \in \mathbb{R}^2 : \|x\| = 1\}$, and to the unit disc $\{x \in \mathbb{R}^2 : \|x\| \leq 1\}$ or $\{x \in \mathbb{R}^2 : \|x\| < 1\}$, respectively.

The *components* of a subset X of S are the equivalence classes of points in X where two points are *equivalent* if they can be joined by an arc in X . The surface S itself, being connected, has only one component.

The *frontier* of X is the set of all points y in S such that every neighbourhood of y meets both X and $S \setminus X$. The frontier F of X separates $S \setminus X$ from X : since $X \cup F$ is closed, every arc from $S \setminus X$ to X has a first point in $X \cup F$, which must lie in F . A component of the frontier of X that is a circle in S is a *boundary circle* of X . A boundary circle of a disc in S is said to *bound* that disc.

There is a fundamental theorem about surfaces, their *classification*. This says that, up to homeomorphism, every surface can be obtained from the sphere $S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}$ by ‘adding finitely many handles or finitely many crosscaps’, and that surfaces obtained by adding different numbers of handles or crosscaps are distinct. We shall not need the classification theorem, but to form a picture let us see what the

¹ Throughout this appendix, ‘connected’ means ‘arc-connected’.